

ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2026

Assignment 4 - Due date 02/10/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github. And to do so you will need to fork our repository and link it to your RStudio.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., “LuanaLima_TSA_A04_Sp26.Rmd”). Then change “Student Name” on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages needed for this assignment: “xlsx” or “readxl”, “ggplot2”, “forecast”, “tseries”, and “Kendall”. Install these packages, if you haven’t done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
#Load/install required package here
library(openxlsx)
library(ggplot2)
library(forecast)
library(tseries)
library(Kendall)
library(readxl)
library(tidyverse)
library(cowplot)
```

Questions

Consider the same data you used for A3 from the spreadsheet “Table_10.1_Renewable_Energy_Production_and_Consumption”. The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. **For this assignment you will work only with the column “Total Renewable Energy Production”.**

```
#Importing data set - you may copy your code from A3
energy_data <- read_excel(path = "../Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx",
                          skip = 12, sheet = "Monthly Data", col_names = FALSE)
```

```

read_col_names <- read_excel(path = "../Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_S",
                             skip = 10, n_max = 1, sheet = "Monthly Data", col_names = FALSE)

colnames(energy_data) <- read_col_names

energy_data$Month <- as.Date(energy_data$Month)
class(energy_data$Month)

## [1] "Date"

renew <- select(energy_data, "Month", "Total Renewable Energy Production")

```

Stochastic Trend and Stationarity Tests

For this part you will work only with the column Total Renewable Energy Production.

Q1

Difference the “Total Renewable Energy Production” series using function `diff()`. Function `diff()` is from package `base` and take three main arguments: * *x* vector containing values to be differenced; * *lag* integer indicating with lag to use; * *differences* integer indicating how many times series should be differenced.

Try differencing at lag 1 only once, i.e., make `lag=1` and `differences=1`. Plot the differenced series. Do the series still seem to have trend?

```

renew_ts <- ts(renew$`Total Renewable Energy Production`, start = c(1973,1), frequency = 12)

renew_diff <- diff(renew_ts, lag = 1, differences = 1)

renew_diff_plot <- autoplot(renew_diff) +
  xlab("Time") +
  ylab("Differenced Value") +
  ggtitle("Differenced Total Renewable Energy Production, lag = 1") +
  theme(plot.title = element_text(hjust = 0.5)) +
  geom_line(color = "blue")

```

Answer: The trend doesn't seem to have a trend because the plot is mostly oscillating around 0 which means the trend is gone after differencing.

Q2

Copy and paste part of your code for A3 where you run the regression for Total Renewable Energy Production and subtract that from the original series. This should be the code for Q3 and Q4. make sure you use assign same name for the time series object that you had in A3, otherwise the code will not work.

```

#create a vector t
nobs <- length(renew_ts)
t <- c(1:nobs)

#linear model for renewable production
renew_lm <- lm(renew_ts ~ t)
summary(renew_lm)

```

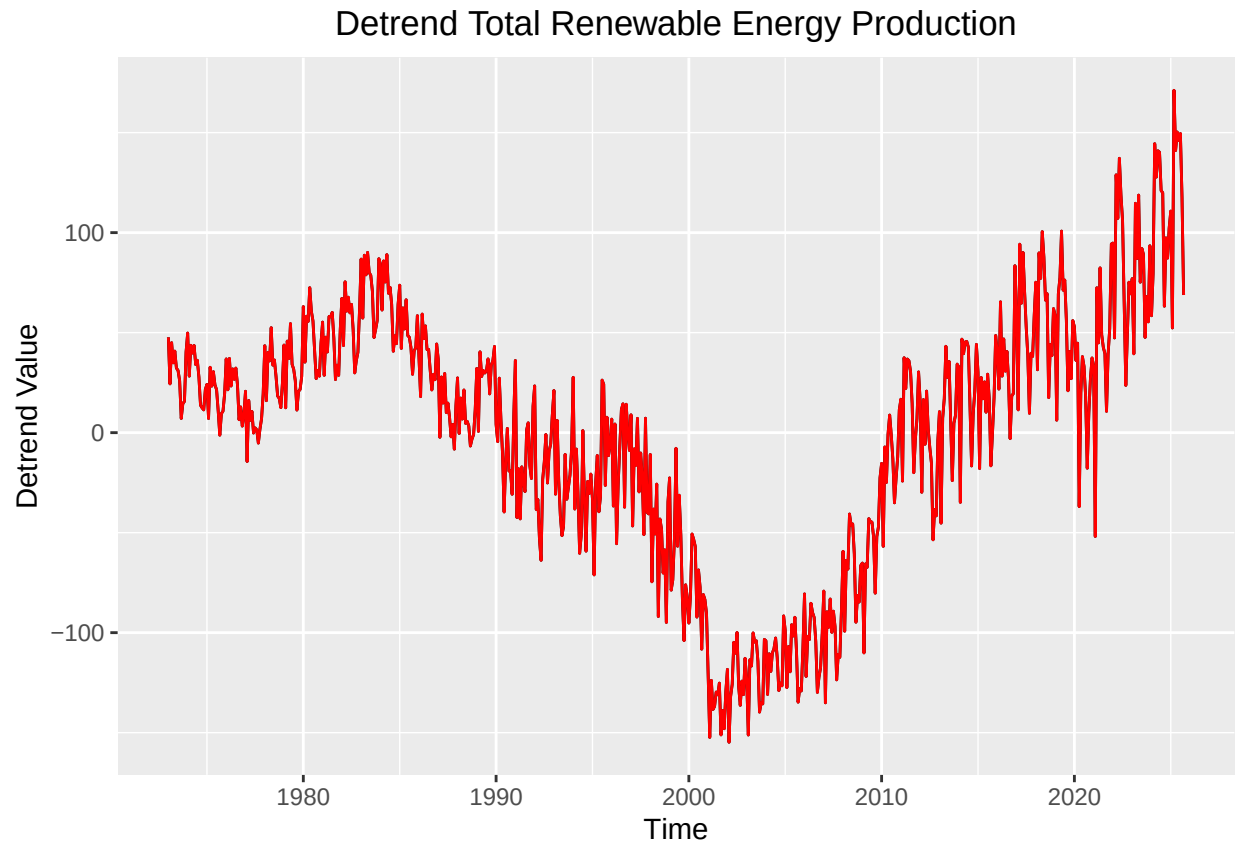
```
##
## Call:
## lm(formula = renew_ts ~ t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -154.81  -39.55   12.52   41.49  171.15
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 171.44868    5.11085   33.55  <2e-16 ***
## t           0.74999    0.01397   53.69  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.22 on 631 degrees of freedom
## Multiple R-squared:  0.8204, Adjusted R-squared:  0.8201
## F-statistic: 2883 on 1 and 631 DF, p-value: < 2.2e-16

int <- as.numeric(renew_lm$coefficients[1])
slope <- as.numeric(renew_lm$coefficients[2])

#detrend
detrend <- renew_ts - (int + slope * t)

detrend_ts <- ts(detrend, start = c(1973,1), frequency = 12)

#plot
renew_detrend_plot <- autoplot(detrend_ts) +
  xlab("Time") +
  ylab("Detrend Value") +
  ggtitle("Detrend Total Renewable Energy Production") +
  theme(plot.title = element_text(hjust = 0.5)) +
  geom_line(color = "red")
renew_detrend_plot
```



Q3

Now let's compare the differenced series with the detrended series you calculated on A3. In other words, for the "Total Renewable Energy Production" compare the differenced series from Q1 with the series you detrended in Q2 using linear regression.

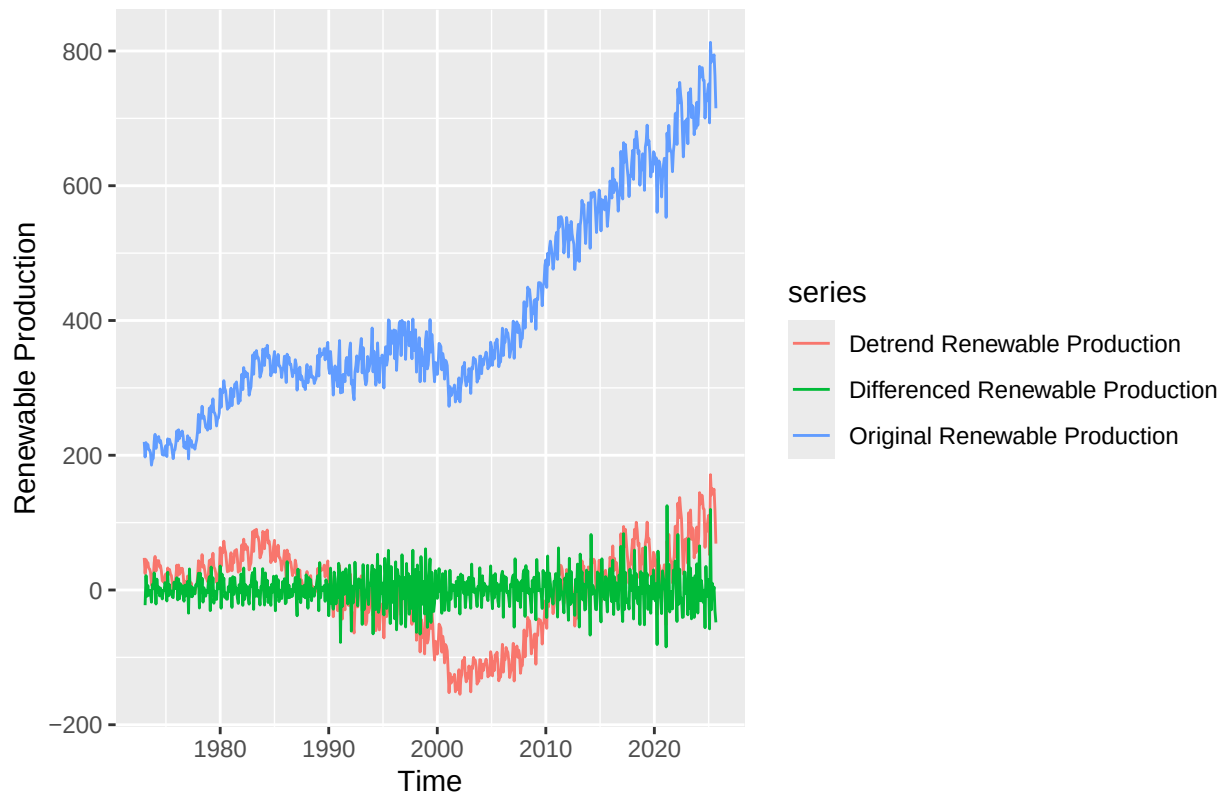
Using `autoplot()` + `autolayer()` create a plot that shows the three series together (i.e. "Original", "Differenced", "Detrended $lm()$ "). Make sure your plot has a legend. The easiest way to do it is by adding the `series=` argument to each `autoplot` and `autolayer` function. Look at the key for A03 for an example on how to use `autoplot()` and `autolayer()`.

What can you tell from this plot? Which method seems to have been more efficient in removing the trend?

```
#compare
diff_vs_detrend_plot <- autoplot(renew_ts, series = "Original Renewable Production") +
  autolayer(detrend_ts, series = "Detrend Renewable Production") +
  autolayer(renew_diff, series = "Differenced Renewable Production") +
  ylab("Renewable Production") +
  ggtitle("Renewable Production Original vs. Detrend vs. Difference") +
  theme(plot.title = element_text(hjust = 0.5))

diff_vs_detrend_plot
```

Renewable Production Original vs. Detrend vs. Difference

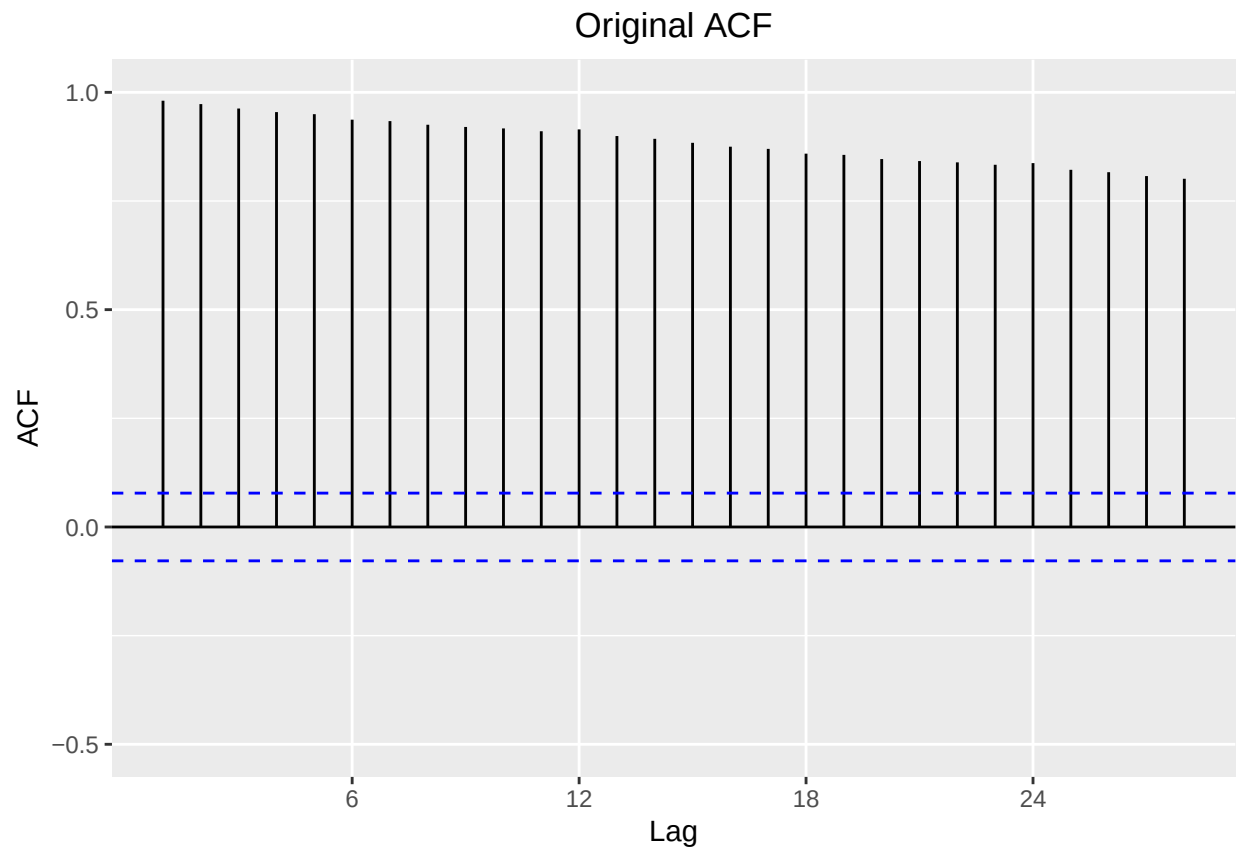


Answer: It seems like detrend and difference plots have distinct differences where detrend has more ups and downs while differenced shows a relatively more stationary plot. With that said, difference is more efficient in removing the trend.

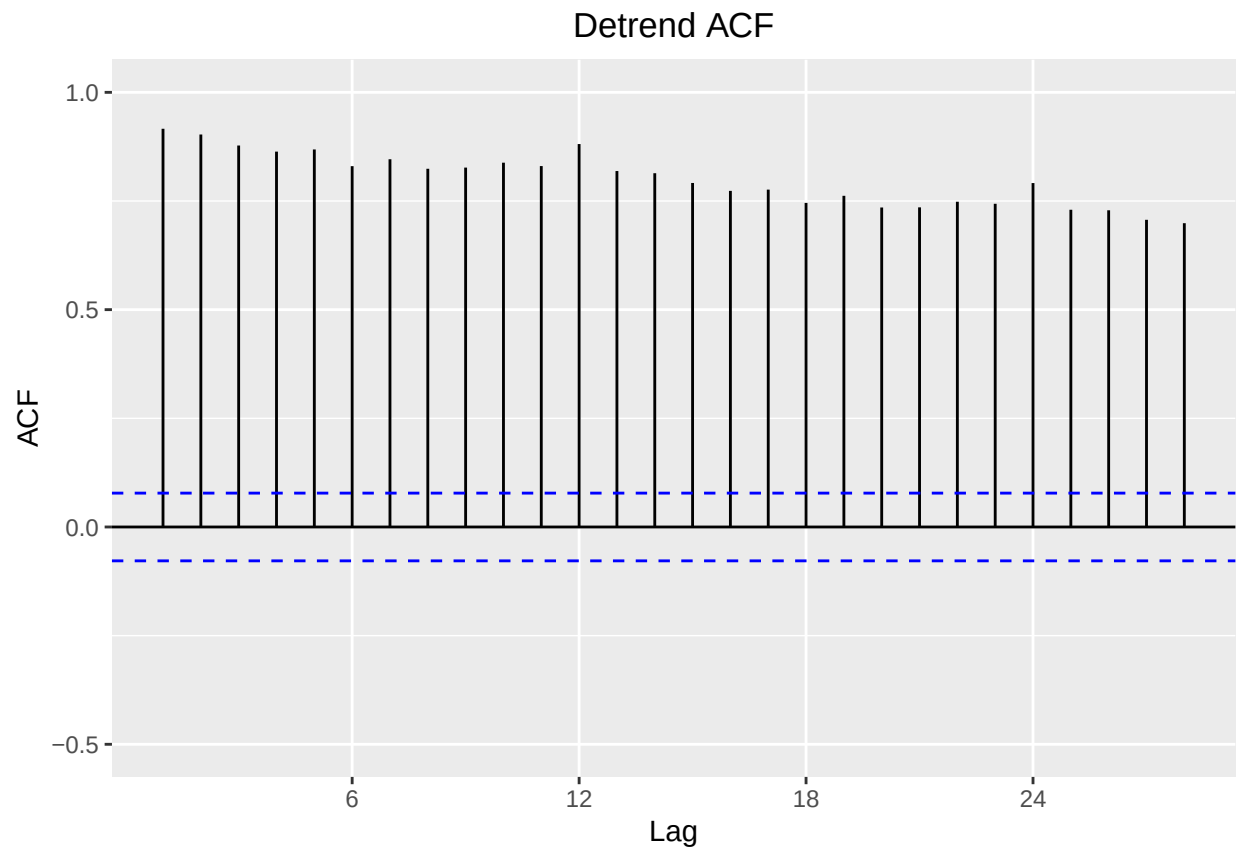
Q4

Plot the ACF for the three series and compare the plots. Add the argument `ylim=c(-0.5,1)` to the `autoplot()` or `Acf()` function - whichever you are using to generate the plots - to make sure all three y axis have the same limits. Looking at the ACF which method do you think was more efficient in eliminating the trend? The linear regression or differencing?

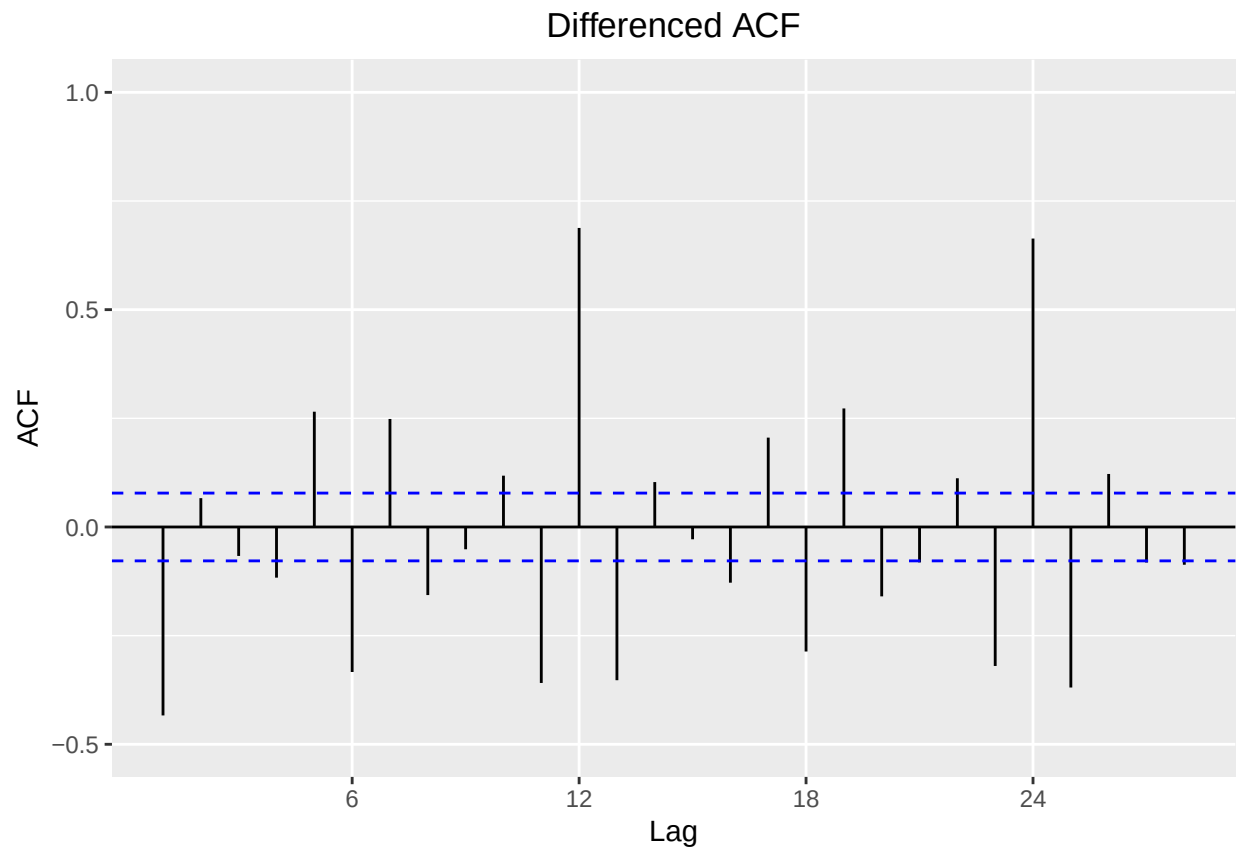
```
original_acf <- ggAcf(renew_ts) +  
  labs(title = "Original ACF") +  
  theme(plot.title = element_text(hjust = 0.5)) +  
  ylim(-0.5, 1)  
original_acf
```



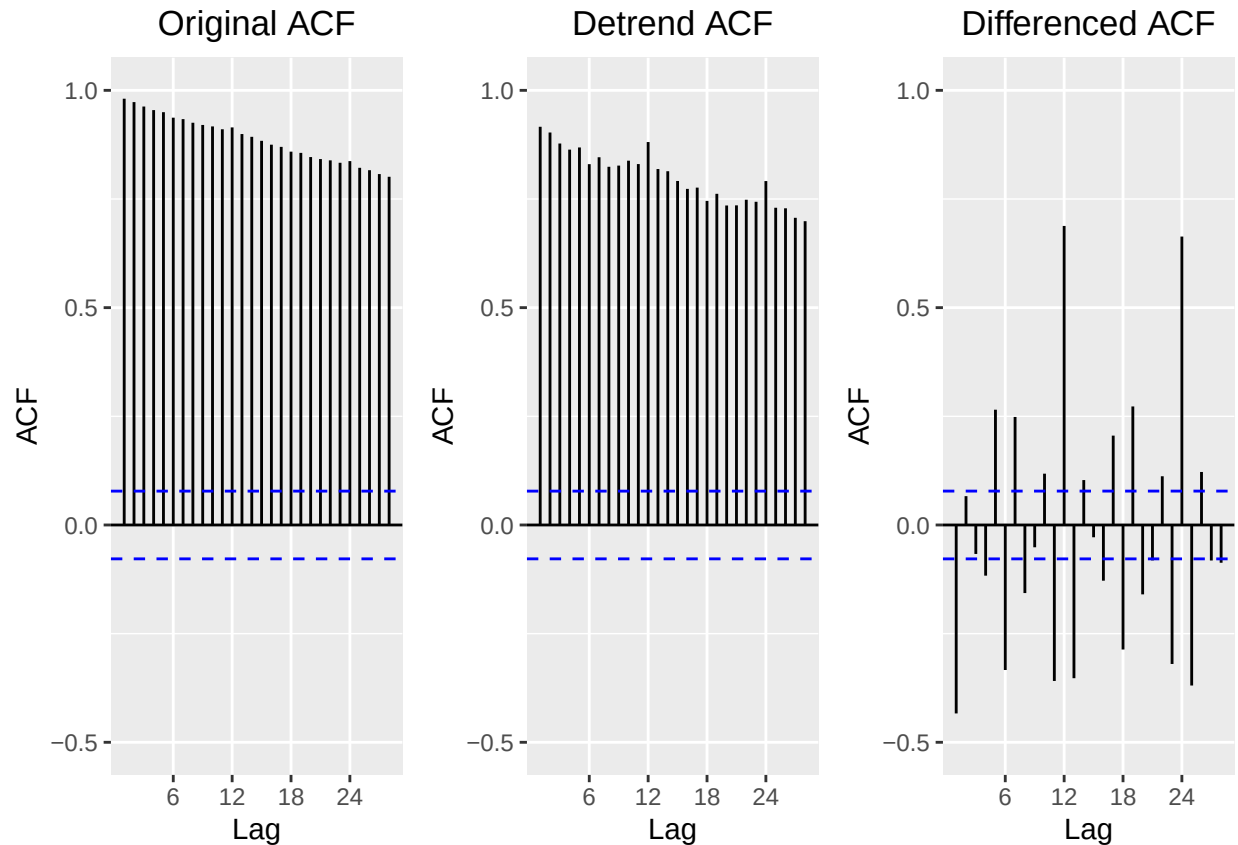
```
detrend_acf <- ggAcf(detrend_ts) +  
  labs(title = "Detrend ACF") +  
  theme(plot.title = element_text(hjust = 0.5)) +  
  ylim(-0.5, 1)  
detrend_acf
```



```
diff_acf <- ggAcf(renew_diff) +  
  labs(title = "Differenced ACF") +  
  theme(plot.title = element_text(hjust = 0.5)) +  
  ylim(-0.5, 1)  
diff_acf
```



```
ACF_comparison <- plot_grid(original_acf, detrend_acf, diff_acf,  
                             ncol = 3, align = "hv")  
ACF_comparison
```



Answer: Differencing seems most effective in eliminating trend compared to detrending. Looking at ACF plots, detrend ACF looks very similar to the original ACF graph where lines gradually declines after many lags, indicating the persistence of influence. However, when looking at differenced ACF, it no longer shows the same pattern. Most correlations are close to zero and the plot looks closer to stationary than detrend ACF.

Q5

Compute the Seasonal Mann-Kendall and ADF Test for the original “Total Renewable Energy Production” series. Ask R to print the results. Interpret the results for both test. What is the conclusion from the Seasonal Mann Kendall test? What’s the conclusion for the ADF test? Do they match what you observed in Q3 plot? Recall that having a unit root means the series has a stochastic trend. And when a series has stochastic trend we need to use differencing to remove the trend.

```
smk_original <- SeasonalMannKendall(renew_ts)
print("Results for Seasonal Mann Kendall /n")
```

```
## [1] "Results for Seasonal Mann Kendall /n"
```

```
print(summary(smk_original))
```

```
## Score = 13083 , Var(Score) = 201135
## denominator = 16379.5
## tau = 0.799, 2-sided pvalue =< 2.22e-16
## NULL
```

```
print("Results for ADF test/n")
```

```
## [1] "Results for ADF test/n"
```

```
print(adf.test(renew_ts, alternative = "stationary"))
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: renew_ts  
## Dickey-Fuller = -1.0247, Lag order = 8, p-value = 0.9347  
## alternative hypothesis: stationary
```

Answer: For the Seasonal Mann-Kendall test, the p-value is 2.2×10^{-15} which is very small and less than 0.05. This means that we reject the null hypothesis. The ADF test has a p-value of 0.9347 rendering we fail to reject the null hypothesis which means that the trend has an unit root and it is stochastic. They match with my observation in Q3 because the plot looks more stationary after differencing which means it is a stochastic trend and differencing effectively remove the trend from the serie.

Q6

Aggregate the original “Total Renewable Energy Production” series by year. You can use the same procedure we used in class. Store series in a matrix where rows represent months and columns represent years. And then take the columns mean using function `colMeans()`. Recall the goal is the remove the seasonal variation from the series to check for trend. Convert the accumulates yearly series into a time series object and plot the series using `autoplot()`.

```
renew_matrix <- matrix(renew_ts, byrow = FALSE, nrow = 12)
```

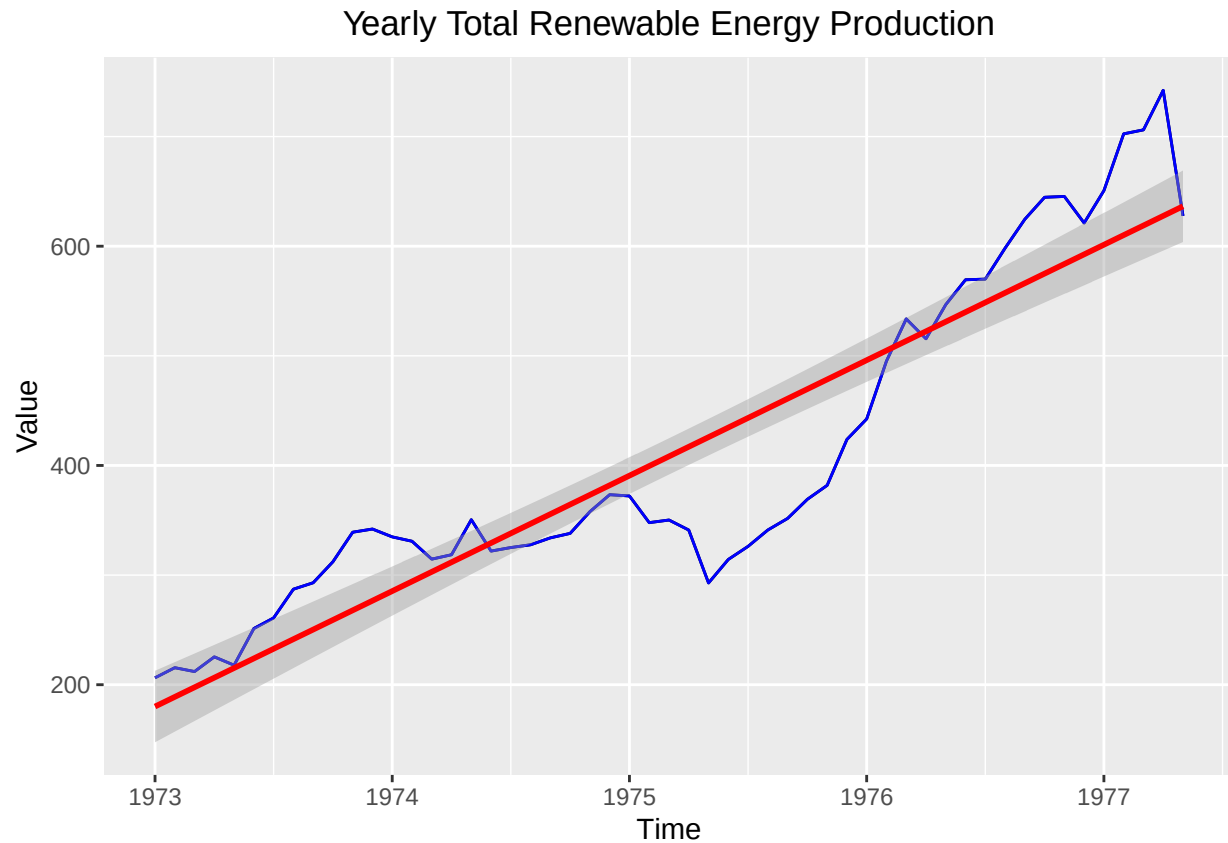
```
## Warning in matrix(renew_ts, byrow = FALSE, nrow = 12): data length [633] is not  
## a sub-multiple or multiple of the number of rows [12]
```

```
renew_yearly <- colMeans(renew_matrix)
```

```
renew_yearly_ts <- ts(renew_yearly, start = c(1973,1), frequency = 12)
```

```
renew_yearly_plot <- autoplot(renew_yearly_ts) +  
  xlab("Time") +  
  ylab("Value") +  
  ggtitle("Yearly Total Renewable Energy Production") +  
  theme(plot.title = element_text(hjust = 0.5)) +  
  geom_line(color = "blue") +  
  geom_smooth(color="red",method="lm")  
renew_yearly_plot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Q7

Apply the Mann Kendall, Spearman correlation rank test and ADF. Are the results from the test in agreement with the test results for the monthly series, i.e., results for Q5?

```
#SMK test
mktest <- MannKendall(renew_yearly_ts)
print("Results for Mann Kendall /n")
```

```
## [1] "Results for Mann Kendall /n"
```

```
print(summary(mktest))
```

```
## Score = 1124 , Var(Score) = 16995.33
## denominator = 1378
## tau = 0.816, 2-sided pvalue =< 2.22e-16
## NULL
```

```
#Spearman
year <- c(year(first(energy_data$Month)):year(last(energy_data$Month)))

sp_rho = cor.test(renew_yearly_ts, year, method = "spearman")
print("Results for Spearman Correlation/n")
```

```
## [1] "Results for Spearman Correlation/n"
```

```
print(sp_rho)
```

```
##  
## Spearman's rank correlation rho  
##  
## data: renew_yearly_ts and year  
## S = 1898, p-value < 2.2e-16  
## alternative hypothesis: true rho is not equal to 0  
## sample estimates:  
##      rho  
## 0.9234801
```

```
#ADF
```

```
print("Results for ADF test/n")
```

```
## [1] "Results for ADF test/n"
```

```
print(adf.test(renew_yearly_ts, alternative = "stationary"))
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: renew_yearly_ts  
## Dickey-Fuller = -1.6789, Lag order = 3, p-value = 0.7037  
## alternative hypothesis: stationary
```

Answer: The result of SMK test has a p-value that is less than 0.05, which means we will reject the null hypothesis indicating the serie follows a trend. The spearman correlation has a result of 0.9234801 which shows a strong positive monotonic association between time and energy production. Lastly, the p-value is 0.7037 for ADF test which means we fail to reject the null hypothesis. Hence, the serie has an unit root and is a stochastic trend. These results resonate with results for Q5.